Extending a Lipschitz map Q -> R_C to a Lipschitz map R_C -> R_C by (R_C,~)-recursion

The Univalent Foundations Program

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Editorial note. The counted statement is the author's RC-extend-Q-Lipschitz (source lines 1073-1076); the proof environment begins two lines below it and the two lines in between are blank, so the pairing is by adjacency.

Editorial note. The retained counter values (chapter 11, section 3, theorem offset 13) make the retained definition print as Definition 11.3.14 and the counted lemma as Lemma 11.3.15, matching the first-edition numbering.

Editorial note. Source line 1079 is a comment in the original text, the author's commented-out uniqueness remark, and stays a comment here; it is not a removal of content.

Editorial note. The definition of Lipschitz maps is retained as context because the statement is phrased with it and the four constructor cases are precisely the four Lipschitz conditions on the constructors of closeness.

Editorial note. No retained line refers to a figure, bibliography entry, included file or non-ASCII character, so the package declares no asset and no bibliography; the only reference outside the retained ranges is the recursion clause declared above.

Editorial scope. One original result of the HoTT Book, counted once: the extension of a Lipschitz map Q -> R_C to a Lipschitz map R_C -> R_C with the same constant.

Allowed standard prerequisites (editorial).

- The higher inductive-inductive Cauchy reals (R_C, closeness). R_C is generated by rat : Q -> R_C together with lim on Cauchy approximations, simultaneously with the family u close_eps v generated by four constructors relating rational points and limits; the accompanying (R_C,close)-recursion principle produces f : R_C -> A from f(rat q), f(lim x), a separated family of relations on A and the four constructor conditions.. The retained proof is one application of that principle, with A := R_C and the relation u ~_eps v := (u close_{L eps} v) set up at source lines 1080-1084.

- Lipschitz maps and Lipschitz constants. f : Q -> R_C is Lipschitz with constant L : Q+ when |q-r| < eps implies f(q) close_{L eps} f(r).. Retained as context at source lines 1056-1069 and consumed in the rational/rational constructor case at line 1107.
- The extra clause of (R_C,close)-recursion. A function f produced by (R_C,close)-recursion satisfies u close_eps v -> f(u) ~_eps f(v); for the Cauchy reals this is equation 11.3.13 of the book.. Source line 1118 quotes exactly this clause to conclude that the extension is Lipschitz with constant L; the clause lies outside the retained span and is declared as a non-load-bearing navigation pointer.

Source reference. Original equation 11.3.13: sources/hott-reals/pinned-original-source.tex, source lines 1050–1050. The counted proof quotes the extra clause of the (R_C,close)-recursion principle at its last line. That clause is stated earlier in the same section and is not part of the counted lemma, so it is a navigation pointer only: the lemma's own author proof is retained line for line and no step of it is replaced by this reference.

Definition 11.3.14. A function $f : \mathbb{Q} \to \mathbb{R}_{\mathsf{c}}$ is **Lipschitz** if there exists $L : \mathbb{Q}_+$ (the **Lipschitz constant**) such that

$$|q - r| < \epsilon \Rightarrow (f(q) \sim_{L\epsilon} f(r))$$

for all $\epsilon : \mathbb{Q}_+$ and $q, r : \mathbb{Q}$. Similarly, $g : \mathbb{R}_{\mathsf{c}} \to \mathbb{R}_{\mathsf{c}}$ is **Lipschitz** if there exists $L : \mathbb{Q}_+$ such that

$$(u \sim_\epsilon v) \Rightarrow (g(u) \sim_{L\epsilon} g(v))$$

for all $\epsilon : \mathbb{Q}_+$ and $u, v : \mathbb{R}_{\mathsf{c}}$..

In particular, note that by the first constructor of $\sim$, if $f : \mathbb{Q} \to \mathbb{Q}$ is Lipschitz in the obvious sense, then so is the composite $\mathbb{Q} \xrightarrow{f} \mathbb{Q} \to \mathbb{R}_{\mathsf{c}}$.

Lemma 11.3.15. *Suppose $f : \mathbb{Q} \to \mathbb{R}_{\mathsf{c}}$ is Lipschitz with constant $L : \mathbb{Q}_+$. Then there exists a Lipschitz map $\bar{f} : \mathbb{R}_{\mathsf{c}} \to \mathbb{R}_{\mathsf{c}}$, also with constant L, such that $\bar{f}(\mathsf{rat}(q)) \equiv f(q)$ for all $q : \mathbb{Q}$.*

Proof. We define $\bar{f}$ by $(\mathbb{R}_{\mathsf{c}}, \sim)$-recursion, with codomain $A :\equiv \mathbb{R}_{\mathsf{c}}$. We define the relation $\frown : \mathbb{R}_{\mathsf{c}} \to \mathbb{R}_{\mathsf{c}} \to \mathbb{Q}_+ \to \mathsf{Prop}$ to be

$$(u \frown_\epsilon v) :\equiv (u \sim_{L\epsilon} v).$$

For $q : \mathbb{Q}$, we define

$$\bar{f}(\mathsf{rat}(q)) :\equiv \mathsf{rat}(f(q)).$$

For a Cauchy approximation $x : \mathbb{Q}_+ \to \mathbb{R}_{\mathsf{c}}$, we define

$$\bar{f}(\mathsf{lim}(x)) :\equiv \mathsf{lim}(\lambda\epsilon.\, \bar{f}(x_{\epsilon/L})).$$

For this to make sense, we must verify that $y :\equiv \lambda\epsilon.\, \bar{f}(x_{\epsilon/L})$ is a Cauchy approximation. However, the inductive hypothesis for this step is that for any $\delta, \epsilon : \mathbb{Q}_+$ we have $\bar{f}(x_\delta) \frown_{\delta+\epsilon} \bar{f}(x_\epsilon)$, i.e. $\bar{f}(x_\delta) \sim_{L\delta+L\epsilon} \bar{f}(x_\epsilon)$. Thus we have

$$y_\delta \equiv f(x_{\delta/L}) \sim_{\delta+\epsilon} f(x_{\epsilon/L}) \equiv y_\epsilon.$$

For proving separatedness, we simply observe that $\forall(\epsilon : \mathbb{Q}_+).\, a \frown_\epsilon b$ means $\forall(\epsilon : \mathbb{Q}_+).\, a \sim_{L\epsilon} b$, which implies $\forall(\epsilon : \mathbb{Q}_+).\, a \sim_\epsilon b$ and thus $a = b$.

To complete the $(\mathbb{R}_{\mathsf{c}}, \sim)$-recursion, it remains to verify the four conditions on $\frown$. This basically amounts to proving that $\bar{f}$ is Lipschitz for all the four constructors of $\sim$.

(i) When u is $\mathsf{rat}(q)$ and v is $\mathsf{rat}(r)$ with $-\epsilon < |q - r| < \epsilon$, the assumption that f is Lipschitz yields $f(q) \sim_{L\epsilon} f(r)$, hence $\bar{f}(\mathsf{rat}(q)) \frown_\epsilon \bar{f}(\mathsf{rat}(r))$ by definition.

(ii) When u is $\mathsf{lim}(x)$ and v is $\mathsf{rat}(q)$ with $x_\eta \sim_{\epsilon-\eta} \mathsf{rat}(q)$, then the inductive hypothesis is $\bar{f}(x_\eta) \sim_{L\epsilon - L\eta} \mathsf{rat}(f(q))$, which proves $\bar{f}(\mathsf{lim}(x)) \sim_{L\epsilon} \bar{f}(\mathsf{rat}(q))$ by the third constructor of $\sim$.

(iii) The symmetric case when u is rational and v is a limit is essentially identical.

(iv) When u is $\mathsf{lim}(x)$ and v is $\mathsf{lim}(y)$, with $\delta, \eta : \mathbb{Q}_+$ such that $x_\delta \sim_{\epsilon-\delta-\eta} y_\eta$, the inductive hypothesis is $\bar{f}(x_\delta) \sim_{L\epsilon - L\delta - L\eta} \bar{f}(y_\eta)$, which proves $\bar{f}(\mathsf{lim}(x)) \sim_{L\epsilon} \bar{f}(\mathsf{lim}(y))$ by the fourth constructor of $\sim$.

This completes the $(\mathbb{R}_{\mathsf{c}}, \sim)$-recursion, and hence the construction of $\bar{f}$. The desired equality $\bar{f}(\mathsf{rat}(q)) \equiv f(q)$ is exactly the first computation rule for $(\mathbb{R}_{\mathsf{c}}, \sim)$-recursion, and the additional condition (11.3.13) says exactly that $\bar{f}$ is Lipschitz with constant L. □